Chapter-10 — 6 hrs

Data Encoding & Modulation. — 15/+5 marks.

Modulation :-

- Modulation may be defined as the process by which source characteristics of a signal called carrier is varied in accordance with the instantaneous value of another signal called modulating signal.

- The frequency of carrier signal is higher than modulating signal.

- The modulation is very benefitial factor in comm^n system some important sides are listed as.

① To design height of antenna.

- We know the heigh of antenna is equal to the quarter of wavelength of the freq. used.

$$\text{ie } H = \lambda/4 = \frac{1}{4}\frac{c}{f}$$

- let us consider A voice signal 2KHz is used to trasmit, for this the antenna height is

$$H = \frac{1}{4}\frac{c}{f} = \frac{1}{4}\frac{3\times10^8 \text{ m/s}}{2\times1000 \text{ Hz}} = 37500\text{m}$$

$$H = 37.5\text{km} \text{ (not in practice}$$

- using modulation technique ie adding carrier signal 5MHz with voice signal, the height becomes

Q. Define Modulation. Explain the use of Modulation.

$$f = 2KHz + 5MHz = 5000000 + 2000 = 5002000$$
$$= 5.002 MHz$$

$$\therefore H = \frac{1}{4} \times \frac{3\times10^8}{5.002\times10^6} m$$

$$= \frac{3\times10^2}{4\times5.002} = 14.99 \simeq 15 m$$

ie $H = 15m$ which is in practical.

(2) To remove Interference
- The freq. range of audio signal is 20 ~~Hz~~ to 20KHz
- In radio, FM broadcasting, If modulation is used all stations broadcast same freq. range ie audio signal which creates interference each other.
- To remove that all stations use diffrent carrier freq. due to which no Interference occure eg 96.1 MHz kantipur 94.8 (radio audio). MHz.

(3) Using modulation schemes noise can be minimized.

(4) Signal transmission rangle also increased ie large coverage of area.

(5) Wireless commn. possible.

2008 Q. What is Modulation? Explain analog modulation. How does it differ from FM.

Types of modulation

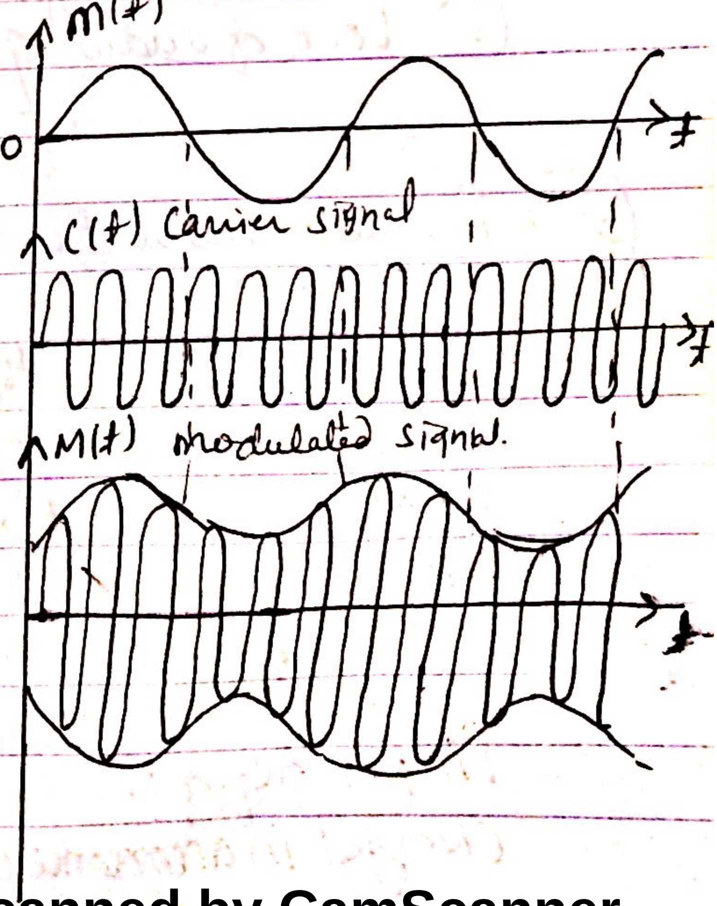
1. Analog modulation
2. Digital modulation.

Analog Modulation :-

It is divided into three groups.

1. Amplitude Modulation.
2. Frequency Modulation. ← } — Angle Modulation.
3. Phase Modulation. ←

① Amplitude Modulation. :-



- When the amplitude of high frequency carrier wave is changed in accordance with the intensity of signal it is called amplitude modulation.

- In AM the amplitude of carrier wave is changed but remaining factor, freq. & Phase remain constant / same.

(SN) AM, FM, PM

The following points are noted in amplitude modulation.

(I) The amplitude of the carrier wave changes according to the intensity of signal.

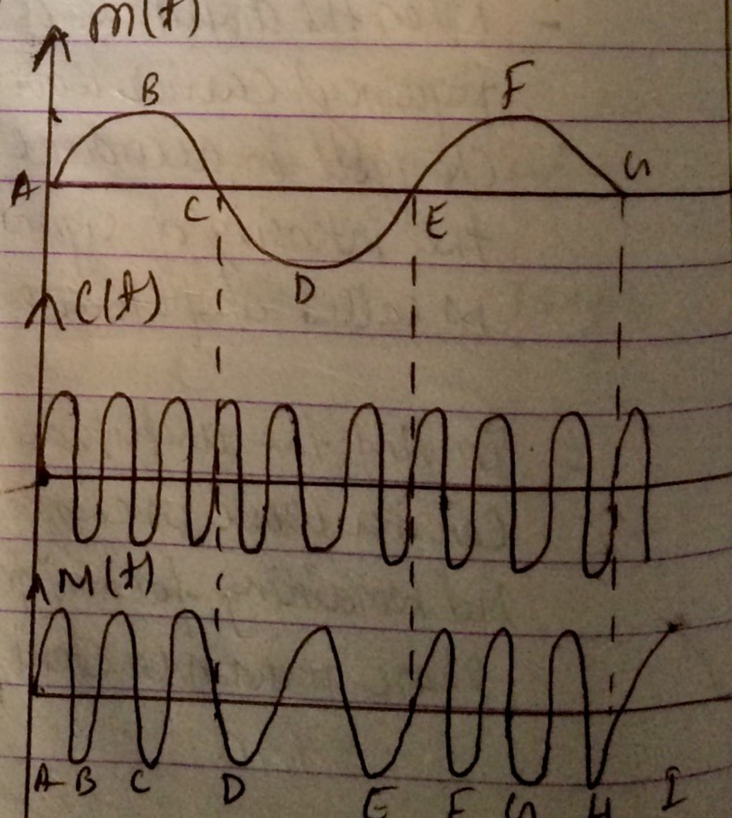
(II) The amplitude variations of the carrier wave is at the signal freq. f_s.

(III) The freq. of amplitude modulated wave remains the same i.e carrier freq. f_c.

Limitation of Amplitude Modulation.

(I) Noisy reception.
(II) Low frequency.
(III) Small operating range.
(IV) Lack of audio quality.

(2) Frequency Modulation :



- When the frequency of the carrier wave is changed in accordance with the intensity of the signal it is called freq. modulation.

- In FM freq. of carrier wave changed in accordance with modulating signal.

- However the amplitude & phase remains same of ~~the~~ carrier freq.

- Following points FM is noted as.

(i) The frequency deviation of FM signal depends on the amplitude of modulation signal.
(ii) The centre frequency is the freq. without modulation or when the modulating voltage is zero.

Advantage of FM over AM.

(1) Operating range quite large.
(2) The efficiency of transmission is very high.
(3) It gives noiseless reception.

Disadvantage of FM.
- FM needs the demodulation which is complex & more expensive than AM.

(3) Phase Modulation (PM)

- When the phase of the carrier wave is changed in accordance with the intensity of the signal it is called phase modulation.
- In this case amplitude & freq. remains constant.

2079 Q. What are different types of analog and digital modulation techniques. Explain AM with detail with waveform.

Digital Modulation.

1. Amplitude Shift Keying (ASK)
2. Frequency Shift Keying (FSK)
3. Phase " " (PSK).

1. Amplitude Shift Keying (ASK)

- ASK is a digital modulation technique defined as the process of shifting the amplitude of the carrier signal between two levels depending on where 1 or 0 is to be transmitted.

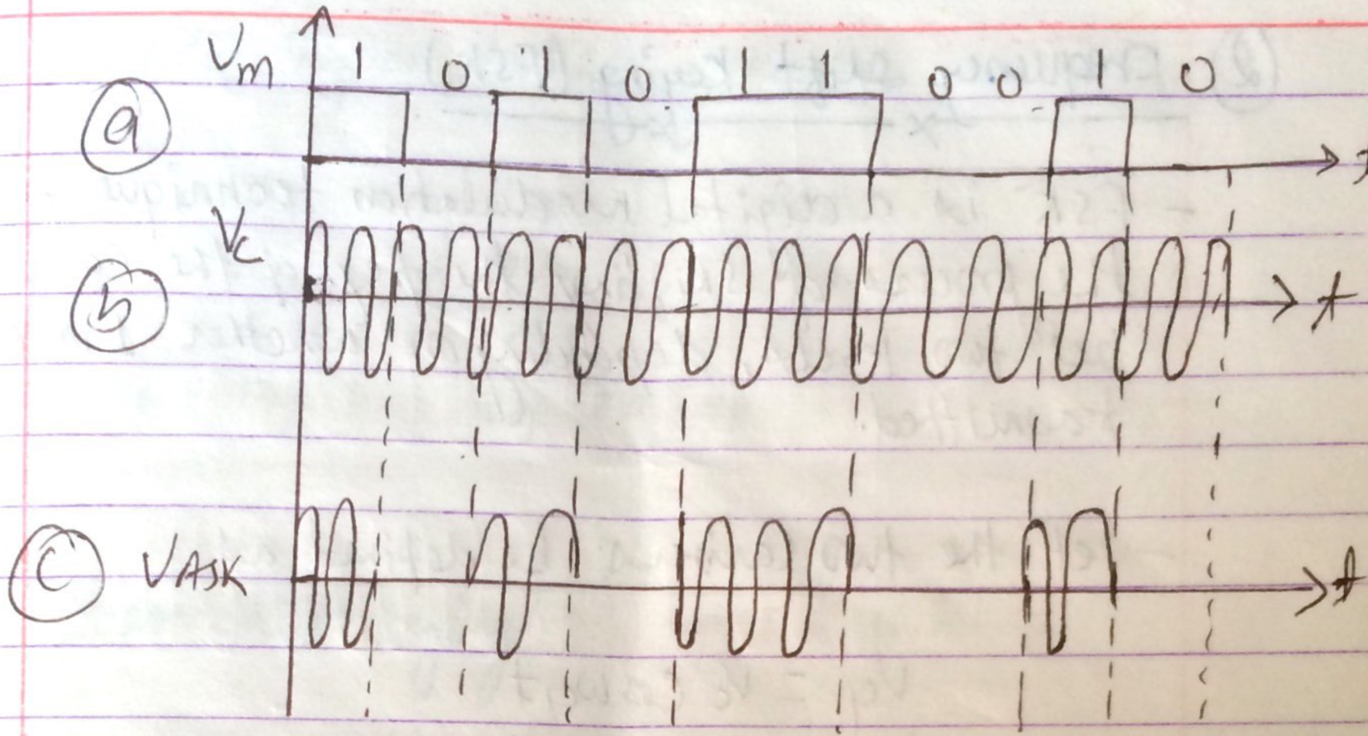
eg, $V_m = V_m$ when symbol is 1 (message signal)
$= 0$ " " " 0

$V_c = V_c \cos \omega_c t$: carrier signal.

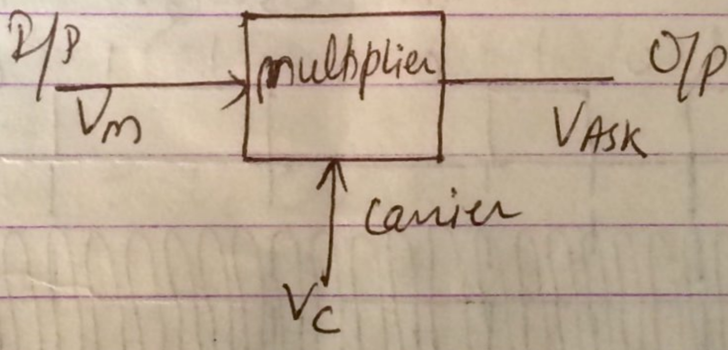
$V_{ASK} = V_m V_c \cos \omega_c t$ at 1
$= 0$ at 0

- The given figure shows the time domain representation of the generation of ASK signal. The digital message ie binary sequence can be represented as message signal.

2017 / 2010 Q. Draw & explain the PCM Block diagram system.



(fig:- time domain representation of generation of ASK signal: (a) message, (b) carrier (c) ASK signal.



[Block diagram of generation of ASK signal]

(2) Frequency shift keying (FSK)

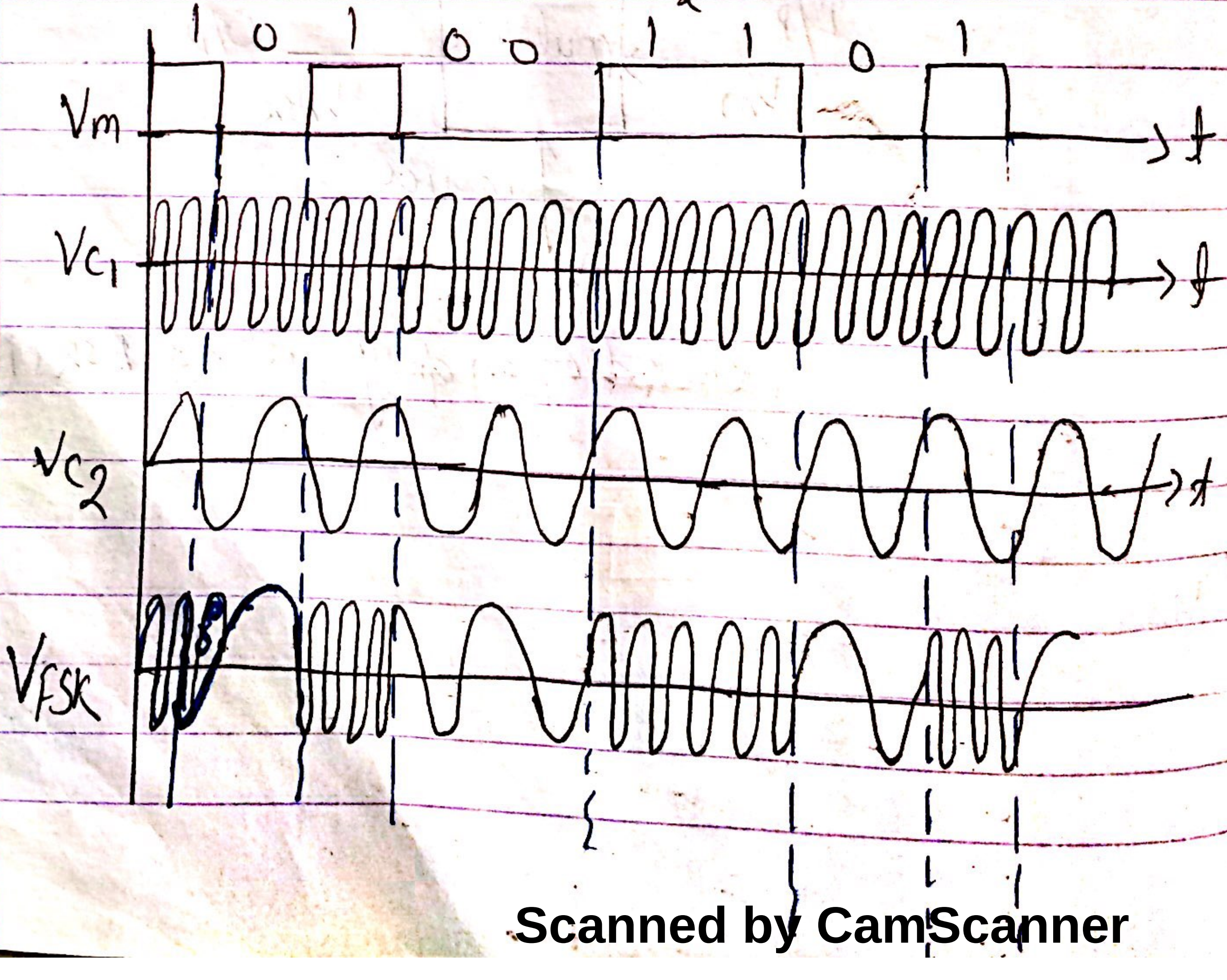
- FSK is a digital modulation technique defined as the process of shifting the freq. of the carrier signal betn two levels, depending on whether 1 or 0 to be trasmitted.

- let the two carriers be defined as

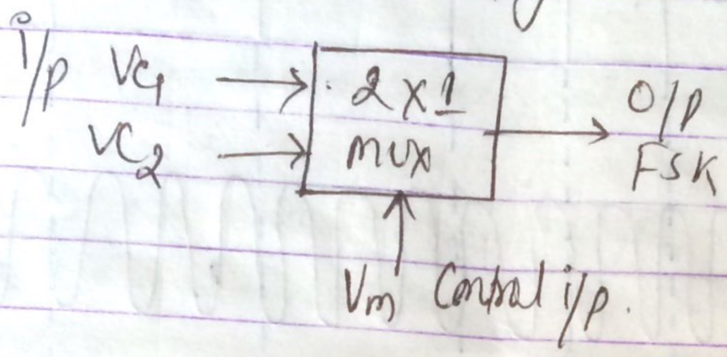
$$V_{C_1} = V_c \cos \omega_{c1} t$$
$$V_{C_2} = V_c \cos \omega_{c2} t$$

- The corresponding FSK signal is defined as

$$V_{ASK} = V_m V_c \cos\omega_{c_1} t \quad \text{at } 1$$
$$= V_m V_c \cos\omega_{c_2} t \quad \text{at } 0.$$



Block diagram of FSK generator



(3) Phase Shift Keying :

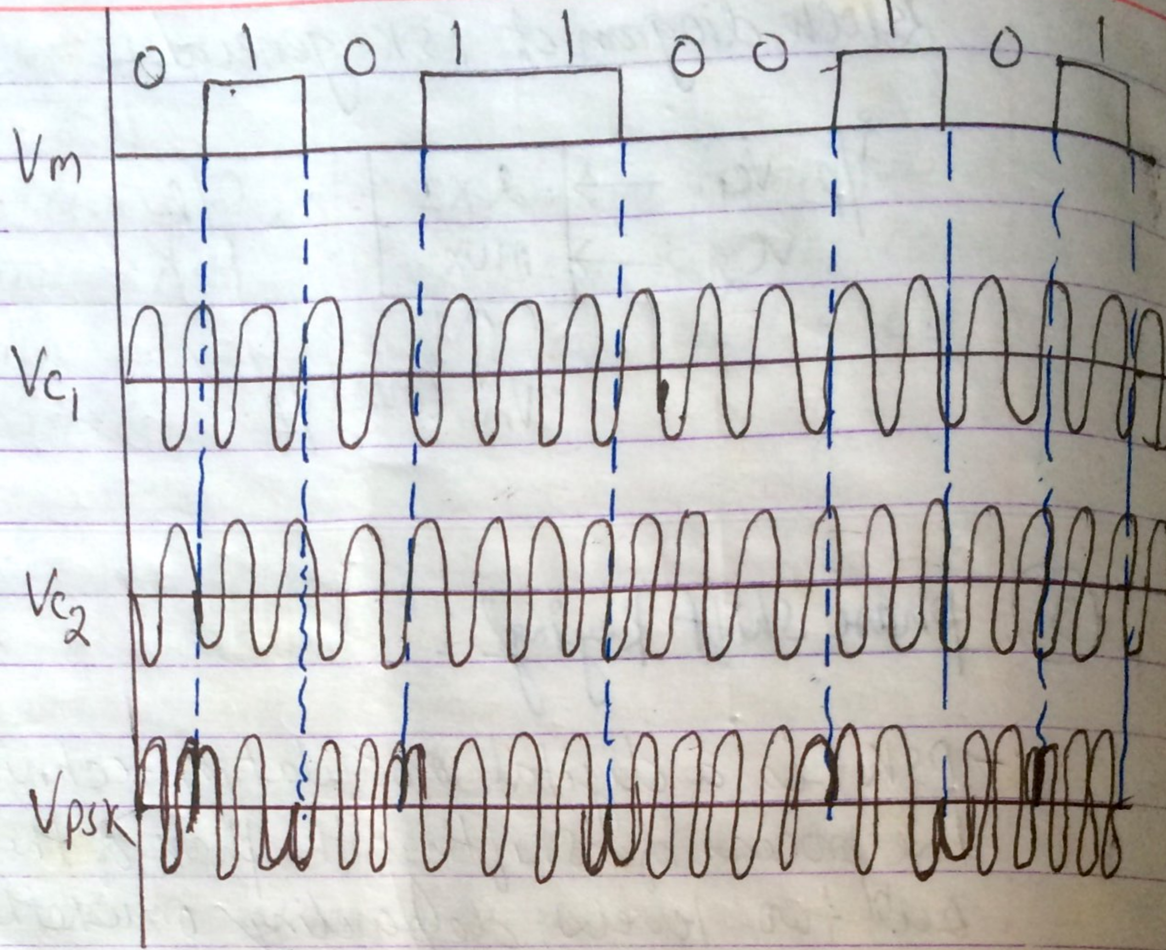
- PSK is a digital modulation technique defined as the process of shifting the phase of the carrier signal betn two levels, depending on whether 1 or 0 is to be trasnitted.

Let the two carriers be defined as

$$Vc_1 = V_c \cos w_c t$$
$$Vc_2 = -V_c \cos w_c t$$

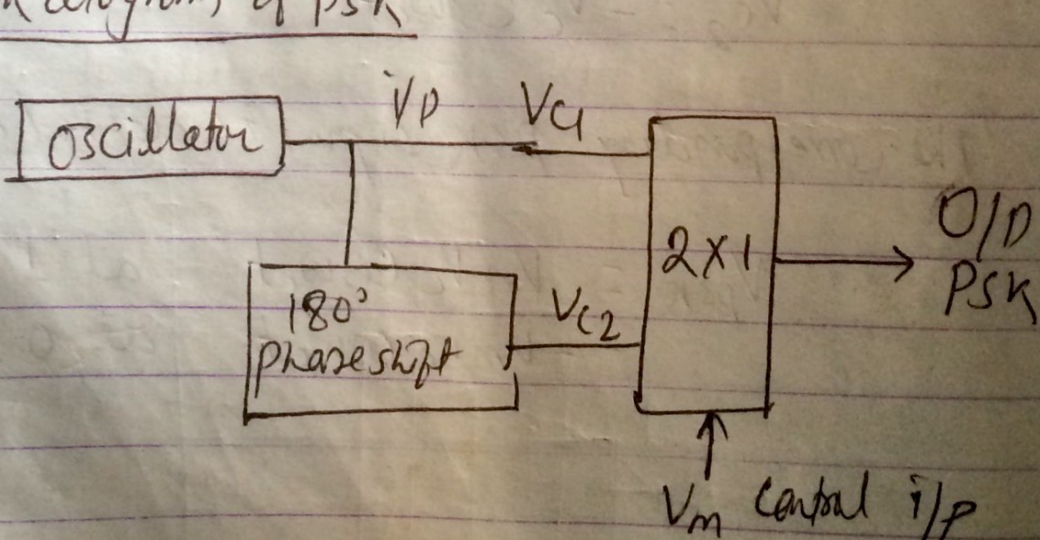
The corresponding PSK signal is defined as

$$V_{PSK} = V_m V_c \cos w_c t \quad \text{at } 1$$
$$= -V_m V_c \cos w_c t \quad \text{at } 0.$$



(fig:- Time domain representation of generation of PSK signal ⓐ message ⓑ carrier with 0° phase shift ⓒ carrier with 180° phase shift & ⓓ PSK signal

Block diagram of PSK



(fig:- Blockdiagram of generation of PSK signal.)

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vvip

Block diagram of PCM system.

- PCM is known as a digital pulse modulation technique.
- It consist of three main parts ie transmitter, transmission path & receiver as shown in figure.

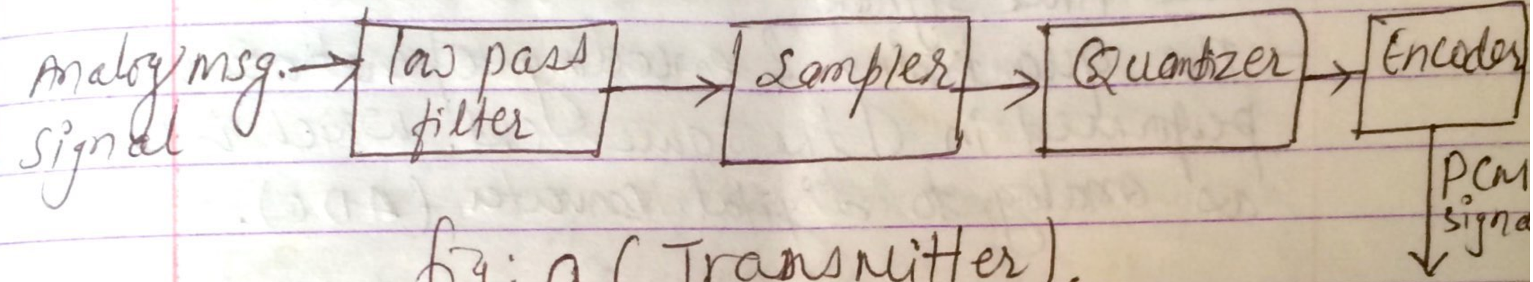


fig: a. (Transmitter).

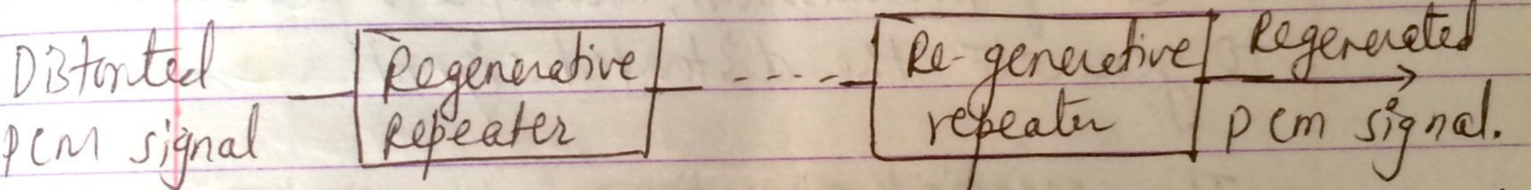


fig: b. (Transmission Media).

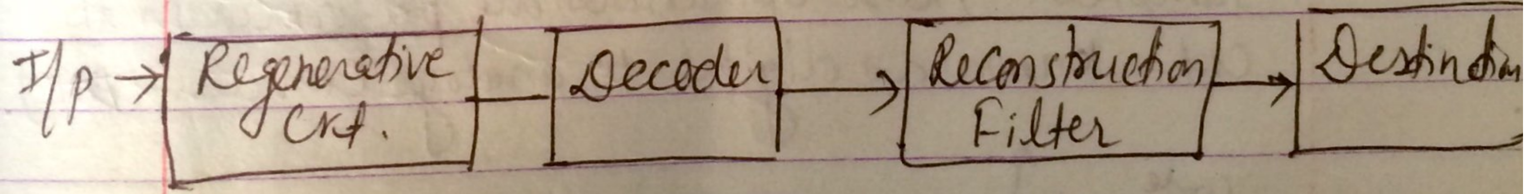


fig:- c. (Receiver)

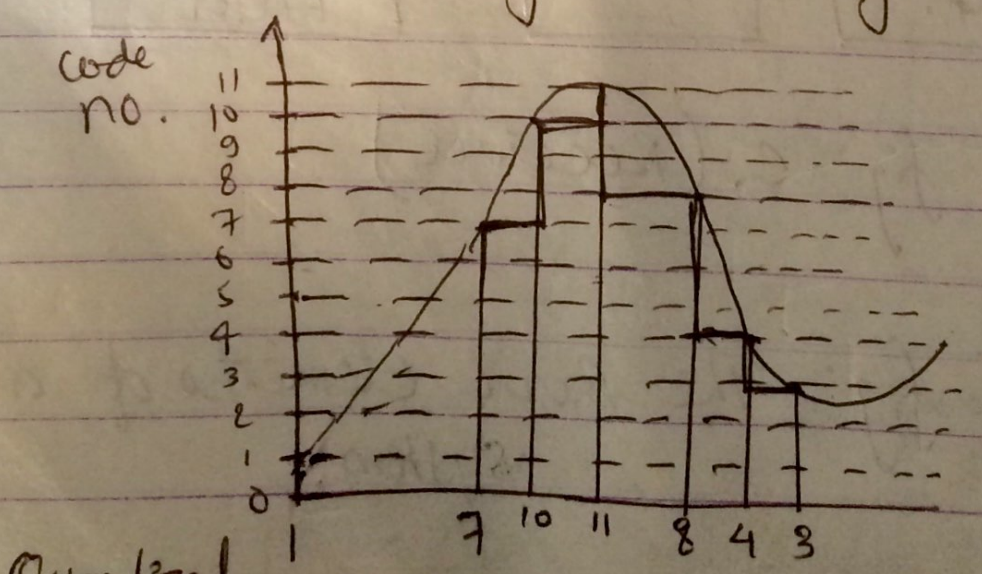
(fig: The Basic elements of a PCM system.)

- The essential operations in the trasmitter of a pcm system are sampling, quantizing & encoding as shown in figure.

- Sampling is the operation in which an analog ie continious signal is sampled according to the sampling theorem resulting in a discrete time signal.
- The quantizing & encoding operations are usually performed in the same ckt. which is known as analog to digital converter (ADC).

- In case of trasmission, multiple repeaters are used to generate the distorted signal.

- The essential operations in the receiver are regeneration of impaired signals ie decoding & de-modulation of the train of quantized samples. These operations are performed in the ckt. known as digital to analog converter (DAC).



(fig: pulse code modulation)

- Quantization assigns fixed digital levels to sampled value.
- Two types of quantization.
- uniform quantization - step size remains same through out the i/p range.
- Non-uniform quantization - non-uniform step size i.e varies according to the i/p signal values

Application of PCM

(1) In telephony (fiber optic cable)
(2) In space communication.

Adventage of PCM

(1) very high noise immunity.
(2) Digital nature of signal.

Disadventage

(1) complex ckt.
(2) require large b/w.

Delta modulation

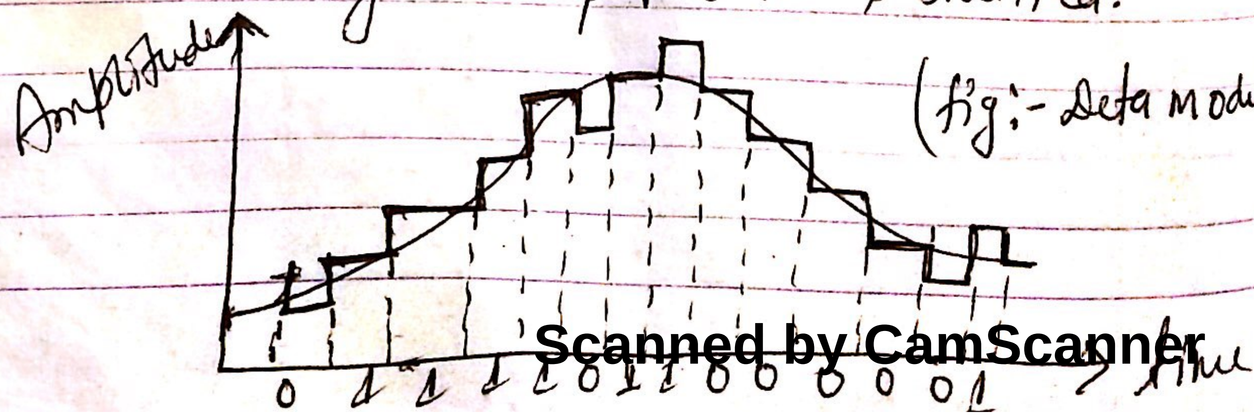
- In PCM, we trasmits all the bits that are used to code sample hence require large bandwidth to overcome this problem Delta modulation is used.

- Delta modulation trasmits only one bit per sample. Here the present sample value is compared with the previous sample value & ~~this result is compared with the previous sample value~~ & this result whether the amplitude is increased or decreased is trasmitted.

- Input signal x(t) is approximated to step signal by the delta modulation.
- Here step size kept fixed. the diffrence bet^n the i/p signal x(t) & staircase approximated signal is confined to two levels ie +Δ & −Δ

- If the difference(Δ) is +ve then approximated signal is increased by one step & '1' is trasmitted

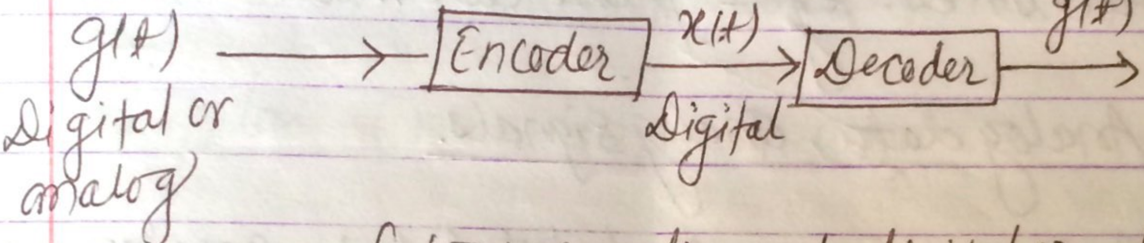
- If the difference(Δ) is -ve, then approximated signal is decreased by one step & '0' is trasmitted.



(fig:- Delta modulation

Encoding :-

- It is the technique of assigning binary value to a signal.
- In case of digital data, encoding is directly done while for analog data, sampling & quantization are necessary before encoding.



(fig :- encoding on to digital signal).

- Depending on data & signals used in transmission diffrent encoding & modulation techniques are adopted as follows.

① **Digital data, digital signal :-**

Digital encoding of digital data is to assign one voltage level to binary one & another to binary zero. eg. line coding.

② **Digital data, analog signal.**

A modem converts digital data to an analog signal so that it can be trasmitted over an analog line. eg ASK, FSK, PSK.

③ Analog data, digital signals:

Analog data such as voice & video are often digitized to be able to use digital trasmission facilities. eg - pulse code Modulation (PCM).

④ Analog data, analog signals.

Analog data are Modulated by a carrier freq^n to produce an analog signal in a different freq^n band which can be utilized on an analog trasmission system.

eg AM, PM, FM.

Imp

Line Coding (Encoding of digital data as digital signal.)

- Binary data can be trasmitted using a no. of different type of pulses. The choice of a particular pair of pulses to represent the symbol 1 & 0 is called line coding.

- After the line coding pulses may be filtered or otherwise shaped to further improve their properties such as spectral efficiency, immunity to ISI (Intersymbol Interface).

⇒ Lines Coding Technique.

- There are various method of representing binary codes with digital waveforms.

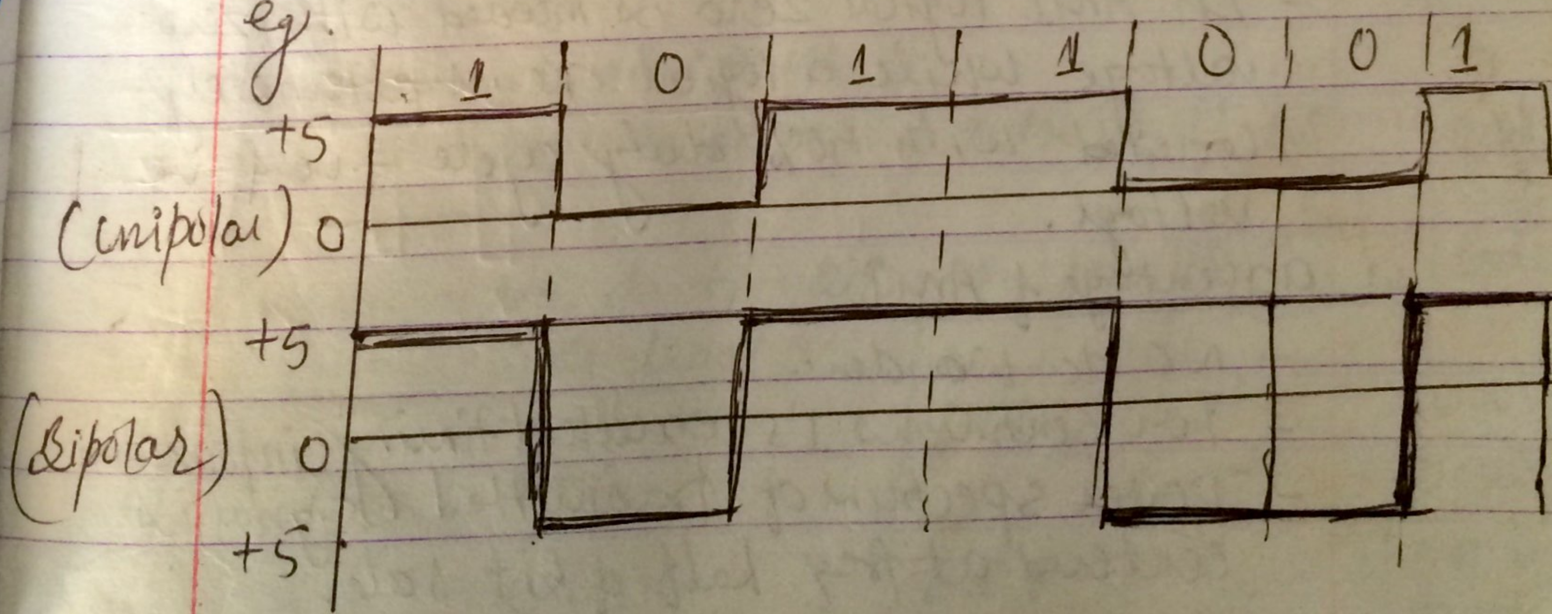
(a) Unipolar (return to zero). RZ

- non zero voltage or current level for 1's & zeroes level.

(b) Bipolar (Non return to zero) (NRZ)

- non-zero positive voltage or current level for 1's & non zero -ve voltage or current for zeroes.

eg.



DC Wonder:

- Continious 1's & 0's signals appears as DC signal of certain level & trasmission lines which are coupled using capacitors & trasformers & blocks DC level so it cause gradual decay of amplitude resulting error in receiver this phenomenon is called DC wonder.

(c) Manchester coding

- Implemented to receive DC wonder.
- 50% of duty cycle of 1st half is given for logical zero & 50% of 2nd of duty cycle is given for logical one.

(d) Alternative Mark Inversion (AMI)

- In AMI logical zero is incoded with zero voltage while a logical one is alternately coded with 50% duty cycle +ve & -ve voltage.

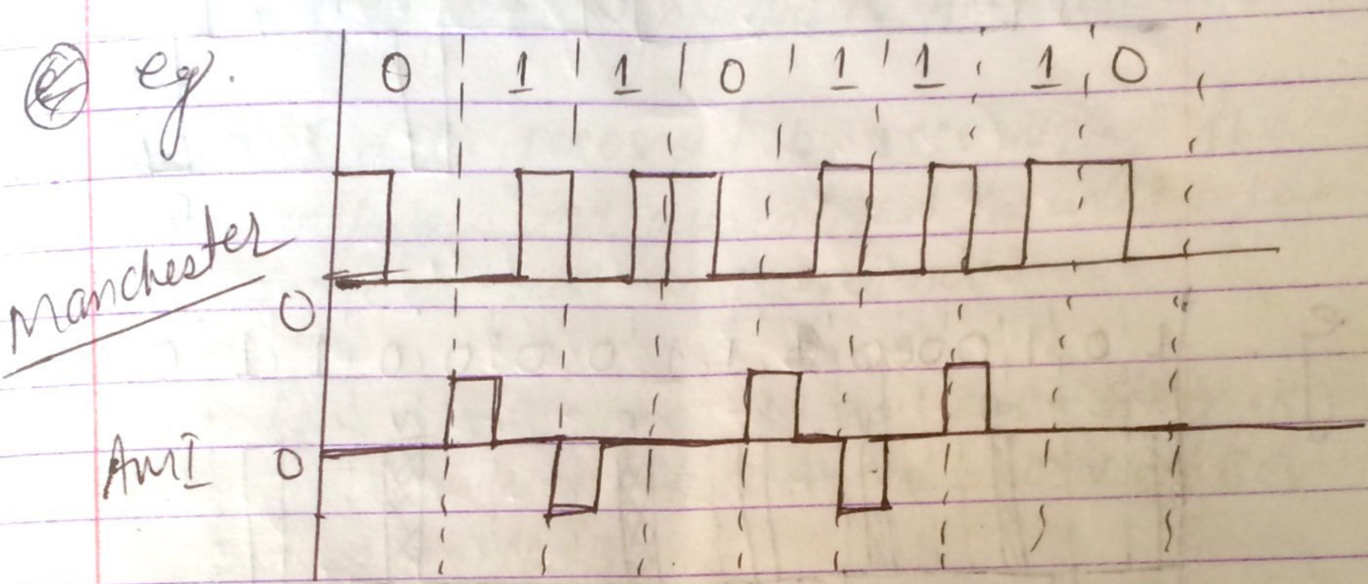
adventage of AMI

- No dc wonder
- For continious 1's excellent timing information
- power spectrum of trasmitted signal is centered at freq. half of bit rate.

disadventage :-

- Fails to provide timing information for long zero transmission.

(d) eg.



(e) HDB3 (High Density Binary coding of order 3)

- each string of four zeroes in the source code is either coded with 000V or B00V.

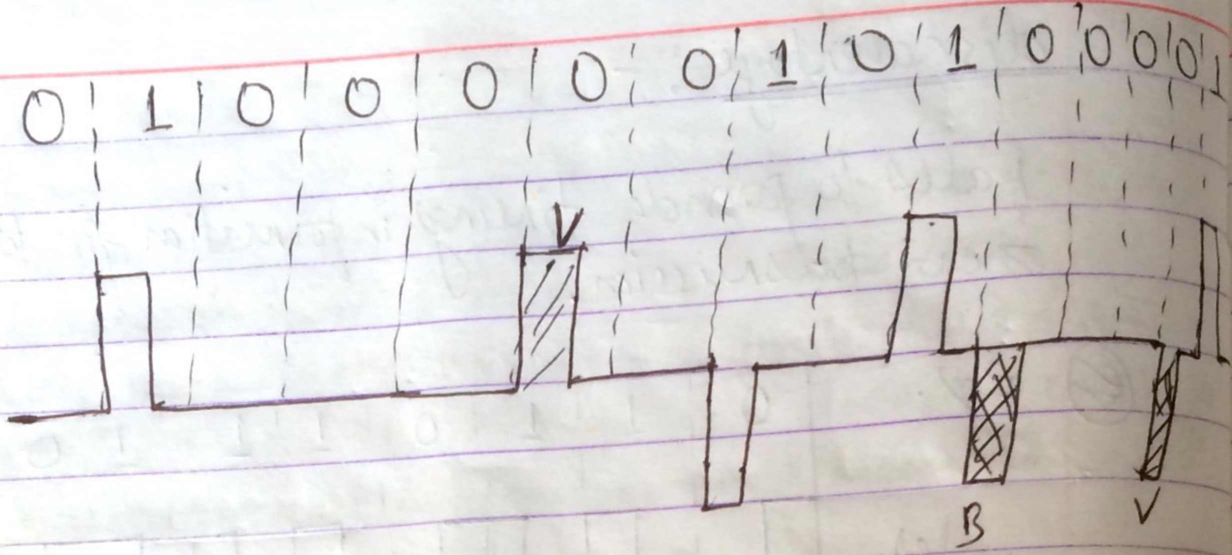
V - violation pulse

B - pulse to keep no. of pulses betn two V. odd.

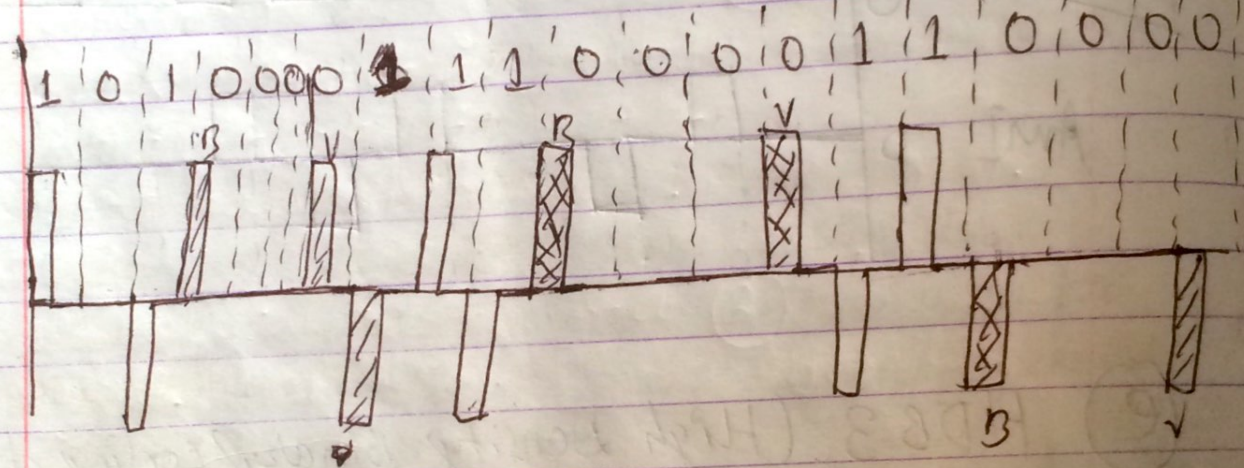
Rule: No. of +/- bits since least V	Pattern	polarity of least pulse	Code
odd	000V	+	$000\overset{+}{V}$
		−	$000\bar{V}$
even	B00V	+	$\bar{B}00\bar{V}$
			$\overset{+}{B}00\overset{+}{V}$

10

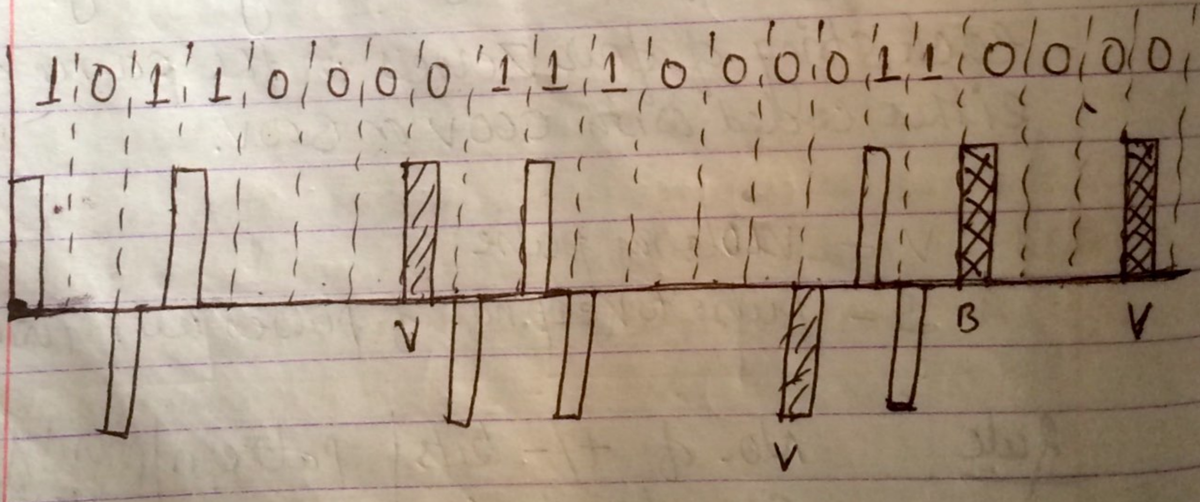
eg


0 1 0 0 0 0 0 1 0 1 0 0 0 0 1
V
B
V

eg.


1 0 1 0 0 0 0 1 1 1 0 0 0 0 0 1 1 0 0 0 0
B
V
B
V
B
V

eg.


1 0 1 1 0 0 0 0 1 1 1 0 0 0 0 1 1 0 0 0 0
V
V
B
V

11

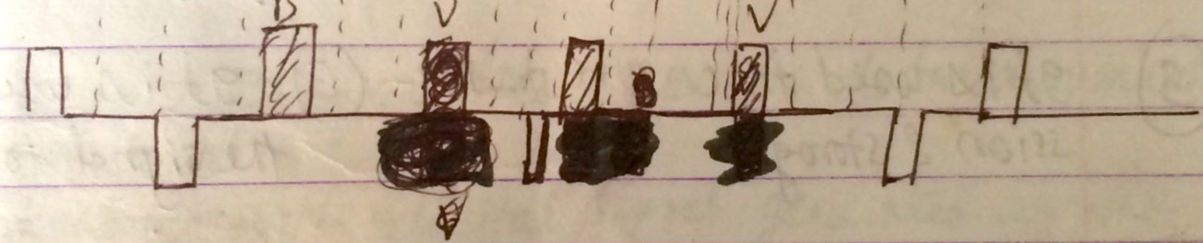
(f) BNZS (Binary N zero subistituter)

- Replace all string 'N' zeroes in the same codes with special source code containing several pulses that properly produce AMI voilations.

- Original data recoverd by recognizing the voilation pulses and pulses that is used to keep no. of pulses bet? V pulses odd.

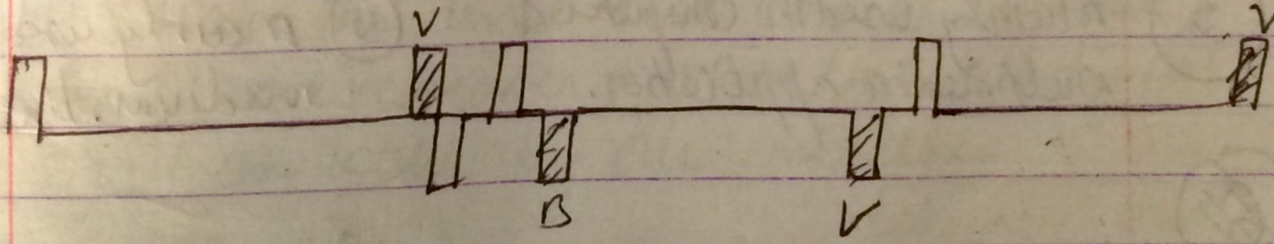
- eg. for N = 3 ie B3ZS each string of 3-zeroes are either encoded 00V or B0V same as in HDB3.

1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 1



- For N = 8, ie B8ZS ie 0000000V 4 B000000V

1 00000000 1 1 000000000 1 00000000



popularly ~~known~~ as (B8ZS)

-

(2) " " FM & AM.

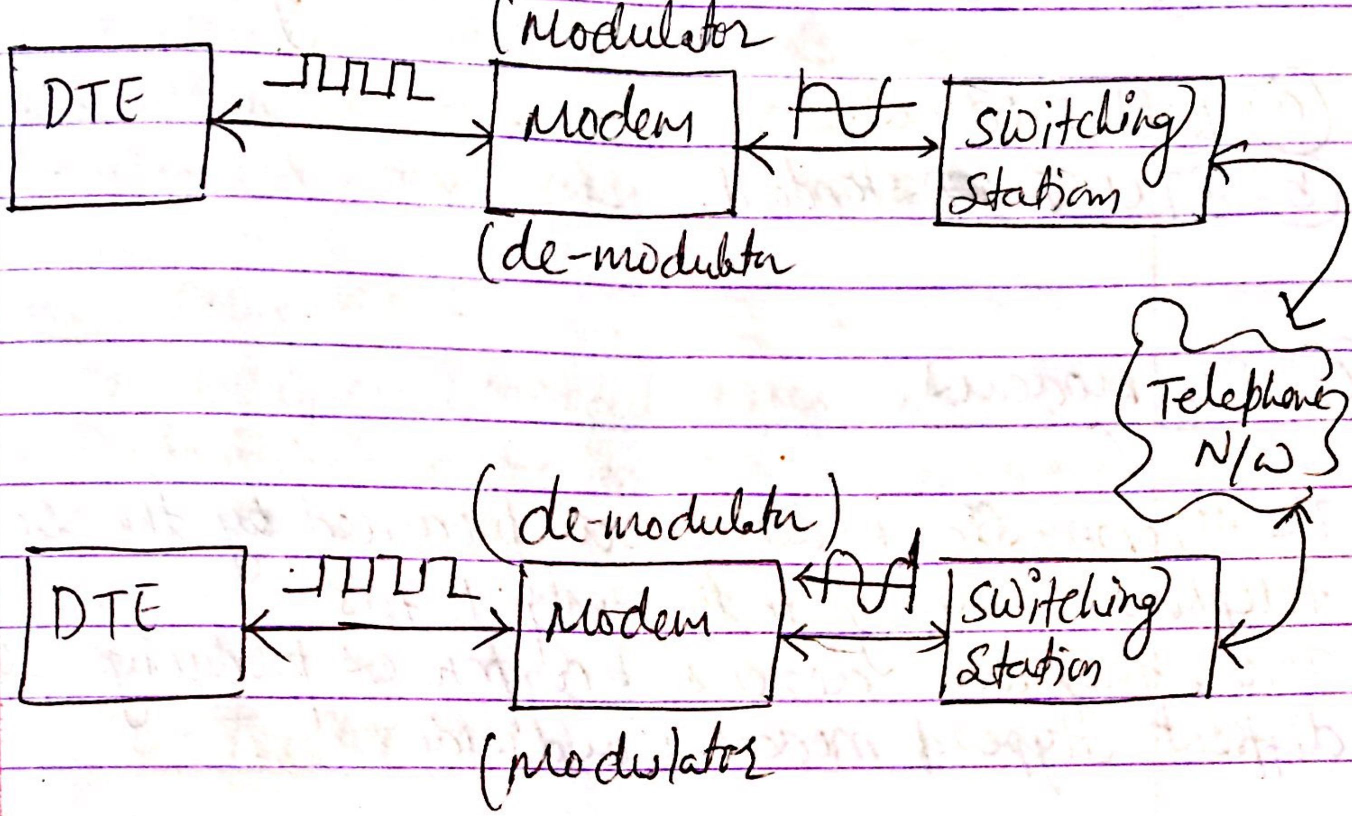
(3) " " multiplexing & non multiplexing.

imp # Difference betn Encoding & Modulation.

Encoding	Modulation.
(1) Encoding is the process by which the data is converted into digital format for efficient transmission or storage.	(1) It is the process to change carrier signal in accordance with message signal.
(2) It is the representation of a signal.	(2) It is about changing of signal.
(3) It is used to ensure transmission & storage.	(3) It is used to send the signal to a long way.
(4) It converts digital or analog data to digital signal.	(4) It converts digital or analog data to analog signal.
(5) mainly used in computer & multimedia application.	(5) mainly used in commu. medium. Like telephone

(6)

Modems :-



(Modem Concept)

- It is most familier type of data communication equipment.
- It converts the digital signal generated by the computer into an analog signal to be carried by a public access phone line.
- It is also the device that converts the analog signal received over a phone line into digital signal usable by our computer.

Modem standards.

Their are two modem standards and they are.

(a) Bell modems.
(b) ITU-T standard.

(a) Bell modems

- The 1st commercial modems were produced by the Bell telephone company in the early 1970's
- Today they are dozens of companies producing 100 of different types of modem world wide.

- The major Bell modems are.

(1) 103 / 113 series

- It is earliest commercial available modem. It operates in full duplex mode over two wire switched telephone line.
- Transmission is asynchronous, FSK modulation
- Data rate is 300bps.

(2) 202 series:

- It operates in half duplex mode.
- Transmission asynchronous, FSK modulation
- Only one transmission freq. is used, due to half duplex

(3) 212 Series

- This series has two speed mode
- 1st, One is 300bps for FSK which is slower
- 2nd, one is 1200bps uses 4-PSK (QPSK) modulation.

(4) 209 series :

- It also operates in full duplex. over 4-wire leased lines.
- Transmission is synchronous. using 16-QAM .
- data rate is 9600bps.

b ITU-T Standard.

- Todays modems are based on this standards.

(1) V. 22 bis :

- The term bis indicates 2nd generation modem of V. 22 series.
- It is two speed modem 1200 & 2400 bps.

(2) V.32 & V.32 bis :

- V32 has combined modulation & encoding technique
- It has data rate of 9600bps.
- The V.32 bis is 1st modem of ITU-T which support 14,400bps. transmission.

(4) V32 terbo

- The terbo refer 3rd version of it has 19,300bps data rate speed.

(5) V.90 & V.92 (High speed modem).

- V.90 has 56.0 Kbps speed
- also called 56K modem
- the uploading 33.6Kbps & Downloading speed is 56Kbps.

- In V.92 uploading/data rate is 48Kbps & downloading speed is 56Kbps.

SW

Higher speed Modem (56 Modem)

- Traditional modems have a limitation on the data rate max.M 33.6 Kbps, so new modems with bit rate 56K bps called 56K modems (ie V.90 & V.92 series) are introduced.

- They are asymmetrical which means diffrent speed of data transmission in downloading & uploading.

- The downloading has max.M bit rate 56Kbps. While uploading has max.M data rate 33.6Kbps.

SW # Multilevel modulation.

- It is the modulation technique to improve the b/w efficiency by trasmitting more info. per symbol.

- This modulation scheme is more sensitive to the channel non-linearities & noise than the binary scheme.
- The main multilevel modulation schemes are

(1) PAM (Pulse Amplitude Modulation)
(2) QAM (Quadrature " ")

- In PAM the amplitude of the individual pulses are varied according to the amplitude of the modulating signals.

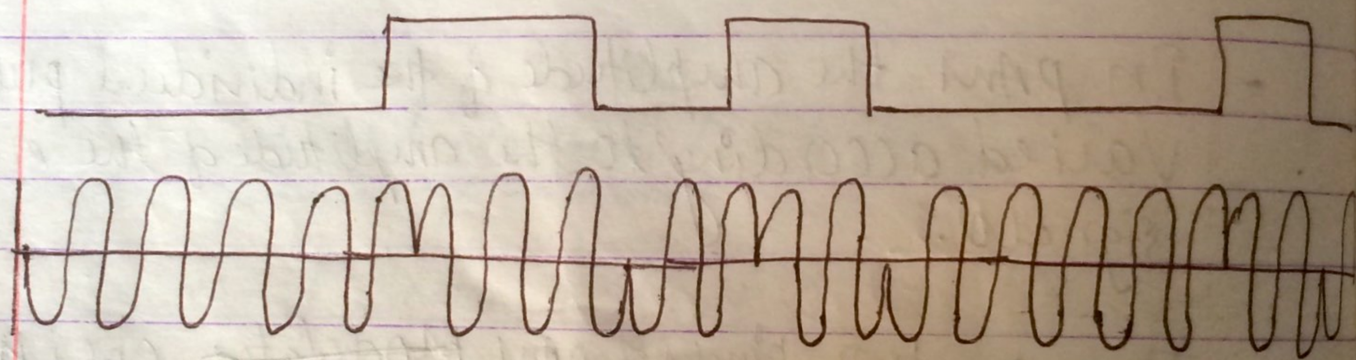
- There are two kinds of PAM ~~Modulation~~. one in which the pulse have the same polarity & other in which the pulses can have both +ve & -ve polarity. of amplitude.

- QAM is a method of combining two amplitude modulated signal into a single channel, thereby doubling the effect B/W.
- QAM is used with PAM in digital systems. especially in Wireless applications.
- QAM has two carrier frequencies with differn phase. one is Q-signal & another is I signal.

DPSK (Differential PSK):-

- It is the type of phase shift keying digital modulation.
- An alternative form of two-level psk is differential PSK.
- In this scheme a binary '0' is represented by sending a signal burst of the same phase as the previous signal burst sent.
- A binary one is represented by sending a signal burst of opposite phase to the preceding one.

0 0 1 1 0 1 0 0 0 1 0



Q.PSK (Four level PSK).

- It is another type of PSK
- In QPSK more efficient use of B/W can be acheived if each signalling element represents more than one bit

- eg). instead of phase shift of 180°, as allowed in BPSK, a common encoding technique uses phase shift of multiples of $\pi/2$ (90°).

- For QPSK

$$s(t) = \begin{cases} A\cos(2\pi f_c t + \pi/4) & 11 \\ A\cos(2\pi f_c t + 3\pi/4) & 01 \\ A\cos(2\pi f_c t - 3\pi/4) & 00 \\ A\cos(2\pi f_c t - \pi/4) & 10 \end{cases}$$